

**CONTINUOUS MANUFACTURING OF
DRUG SUBSTANCES AND DRUG PRODUCTS**

Q13 Annexes Draft

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PART II: ANNEXES

ANNEX I: CONTINUOUS MANUFACTURING OF DRUG SUBSTANCE FOR CHEMICAL ENTITIES

1. INTRODUCTION

For drug substance manufacture, CM often provides specific control of factors affecting transformations (e.g., surface area to volume ratios, degree of mixing, or heat transfer). Therefore, the reaction and separation processes can be controlled more precisely and efficiently. This unique aspect enables synthetic routes that may not be possible in batch manufacture and more consistent production of the drug substance with high purity. This annex highlights key aspects or points to consider for CM processes of the drug substance for chemical entities, with focus on CM specific elements of a control strategy to reduce or eliminate impurities and achieve the desired quality of the drug substance, based on the principles described in the main Q13 guideline.

This annex covers CM processes of the drug substance that generally includes process steps consisting of continuous synthesis (e.g., flow chemistry), workup (e.g., extraction), and purification (e.g., continuous crystallisation). These drug substance manufacturing processes can have different levels of integration of continuous process steps: a full integration of all process steps into a single end-to-end CM process of the drug substance (Figure 1A) or partial integration of multiple continuous process steps with the remaining process step(s) operating in a batch mode (e.g., crystallisation) (Figures 1B and 1C).

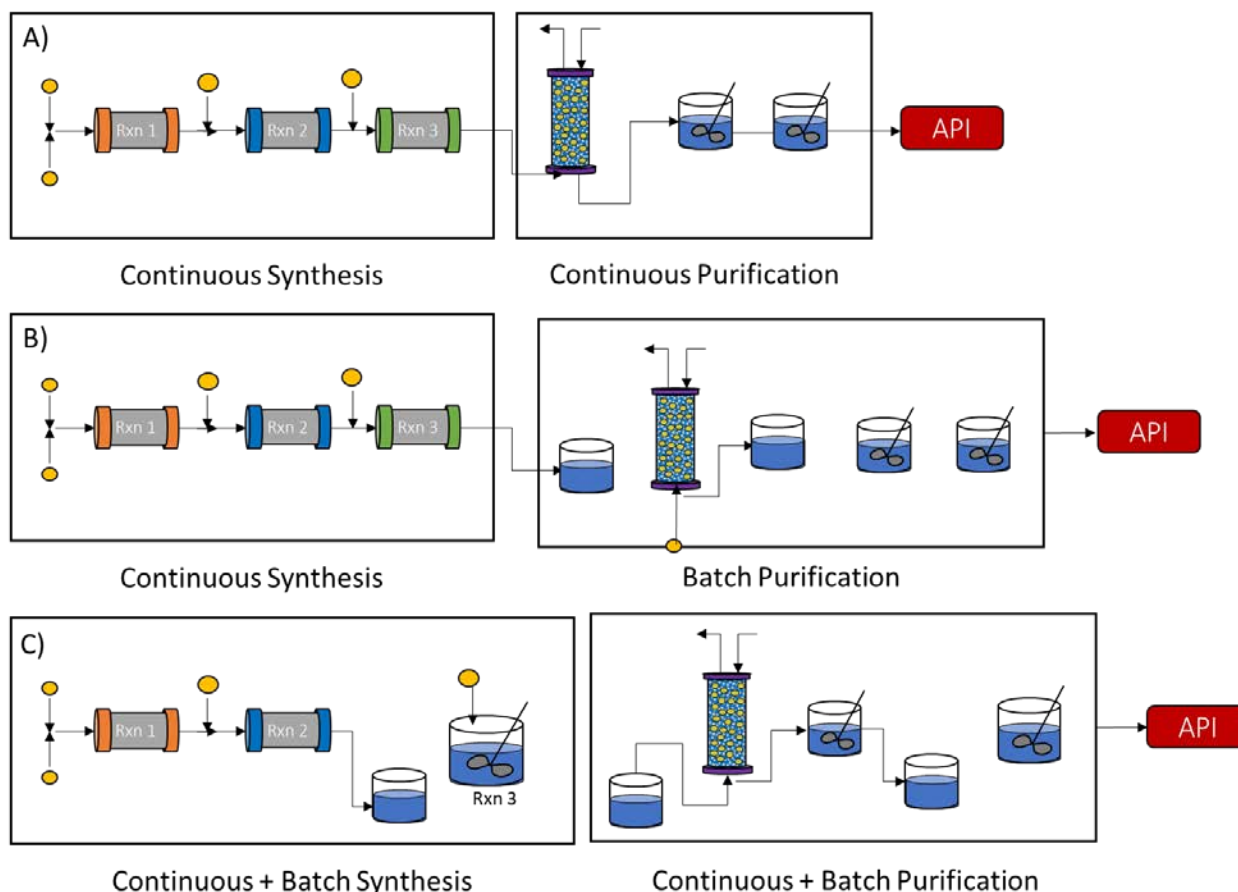


Figure 1: Schematic diagrams for CM processes of drug substances (A) Fully integrated CM process from starting materials to the final drug substance; (B) partially integrated process that includes multiple integrated continuous flow synthesis steps and batch purification steps; and (C) partially integrated process that includes both integrated continuous flow and batch synthesis steps as well as integrated continuous and batch purification steps. Yellow circles represent starting materials, raw materials, intermediates, and other materials.

2. BATCH SIZE

Based on the batch definition given in ICH Q7 (Section 2.2), in processes that include both integrated continuous and batch process steps, the size of a batch should be established by considering the interfacing batch operations. If a batch purification and/or crystallisation is at 100L scale, and this step takes 24 hours to accumulate material from the CM process, the relation of the runtime of the CM part of manufacture with this volume should be described. For example, it may be possible, with justification and supporting data, to establish a validated range of operation 100L to 2400L (isolated in 100L lots) with a run time of one day to 24 days for the CM process, respectively.

3. CONTROL STRATEGY

A science and risk-based approach should be used to assess the origin, formation, and fate of impurities throughout the synthesis and to evaluate the impact of each transformation on the material quality attributes throughout the entire process including integrated continuous process steps. This knowledge should then be used to develop an appropriate control strategy that assures that CQAs of the final drug substance, including an impurity profile and, if applicable, physicochemical properties (e.g., particle size distribution and polymorphic form), are met for collected output materials. The control strategy should also ensure that the non-conforming materials are managed appropriately.

The following aspects should be considered during the development of a control strategy for drug substance CM processes.

3.1. Process Chemistry and Impurities

The origin, formation, and fate of impurities should be assessed by considering the chemistry and process conditions for impurity formation and the capability of process steps for impurity removal. The consequences of telescoping multiple transformations together into a continuous process step(s) should be evaluated with respect to impact on the ability of the process to purge impurities as well as to achieve the required purity of in-process material(s) and final product. This evaluation should consider a range of operating conditions based on a risk-based approach (e.g., possible changes or variations in stoichiometry, temperatures, and flow rate). This evaluation should also consider realistic manufacturing operations caused by start-up, extended operation, and/or process pauses.

The impurity assessment may incorporate relevant elements of prior knowledge and/or information generated from well-designed experiments, modelling, and simulations. The results of this assessment should be used to support the development of control strategy for impurities, such as establishing specifications for proposed starting materials and, if applicable, intermediate(s), parametric controls (e.g., setting appropriate limits for CPPs), PATs for real-time monitoring and controls, and/or end-product testing in the drug substance specification.

3.2. Equipment Design and Integration

Equipment design is an essential element of a control strategy for CM processes of the drug substance. For example, reactors, pumps and flow controllers, crystallisers, and ancillary equipment should be designed, constructed, and assembled with considerations of system requirements on delivering desired chemistry, transformation, and quality of the output material during the period of continuous operation. These considerations should include, but not be limited to, achieving the desired mixing characteristics, residence time, and residence time distribution, as well as heat and mass transfer of a reactor. The reliability, accuracy, and consistency of pumps, flow controllers, and temperature and pressure control systems may also be important considerations.

Where required, CM processes can deliver more precise performance as each process step or equipment can be designed to give the appropriate conditions to maximise the desired transformations (e.g., intermediates) and minimise or eliminate undesired transformations (e.g.,

impurities) or ensure the consistency of drug substance physicochemical attributes (e.g., particle size). This higher level of control by process design may lead to fewer purification steps to achieve a target drug substance quality, in comparison to an equivalent batch manufacturing process.

The design of a system and its integration (see Section 3.1.4 of the main Q13 Guideline for general considerations) is also important in determining the overall process performance and the quality of output materials, as it defines the RTD of material flowing through the process steps and forms the basis of making operational decisions on material diversion and collection. In some drug substance CM processes, surge tanks are used and integrated in between process steps to help manage fluctuations during start-up, process pauses, shutdown, or changes in material flow rate. The design of these surge tanks and their integration with other process steps will impact the RTD of the system.

The level of detail to be provided regarding equipment design and integration should be commensurate with its impact on the state of control operation in the context of the overall control strategy. For example, in the case of a CM process of drug substance where the parametric controls are used as a primary in-process control, the critical aspects of equipment design should be evaluated together with parametric control elements (i.e., process parameters and their range) to understand their impact on the final drug substance quality based on the principles of QbD described in ICH Q8.

As in all manufacturing processes, consideration should be given to maintenance and verification of the cleanliness of the equipment. The equipment maintenance and cleaning strategy should be an essential part of considerations for determining the maximum runtime or extending runtime for a CM process.

3.3. Process Monitoring and Controls

CM processes of the drug substance can have slow process dynamics (e.g., hours) or fast process dynamics (e.g., seconds), and therefore the residence time of different process steps within a CM process can vary significantly. For example, due to differing reaction kinetics, a chemical reaction in Step 1 may have a residence time of less than a minute, but a chemical reaction in Step 2 may have a residence time of over an hour. The equipment and operation can be designed to allow these two steps to be coupled and integrated by providing a larger volume for the step with slower dynamics (and therefore longer residence time) for the same input flow rates. In addition, depending on the nature of transformations, different flow characteristics (e.g., a plug flow or well mixed system) may be required to achieve the desired conversion, yield, or quality of materials.

Given the above scenarios, process monitoring and sampling strategy (e.g., sampling or measurement frequency) should be selected to suit the process dynamics or mixing characteristics of the different process steps or operations. For instance, a process step that exhibit slow dynamics (e.g., a surge tank) may provide an opportunity for using PAT tools such as on-line or at-line HPLC to monitor directly quality attributes of materials (e.g., purity or impurities).

The criteria and procedures for the process start-up, shutdown, and pauses or returning back to a state of control operation should take into consideration (possibly) different response times of

different process steps and the system to transient events. Under these circumstances, it may be suitable to adjust (e.g., increase) the frequency of monitoring to ensure the CM system under a state of control before starting to collect output materials.

Other general considerations for process monitoring and control are provided in Section 3.1.5 of the main Q13 Guideline.

ANNEX II: CONTINUOUS MANUFACTURING FOR DRUG PRODUCT

1. INTRODUCTION

This annex provides considerations specific to the CM of solid oral drug products (i.e., tablets and capsules) based on the principles described in the main Q13 guideline. In particular, it covers CM systems that involve a full integration of all powder processing steps into a single end-to-end drug product manufacturing process or a partial integration of multiple continuous powder processing steps with the remaining process step(s) operating in batch mode (e.g., tablet coating). Examples of these CM systems include continuous direct compression, continuous dry granulation, and continuous wet granulation systems. Figure 1 shows full and partial integrations for a continuous direct compression system.

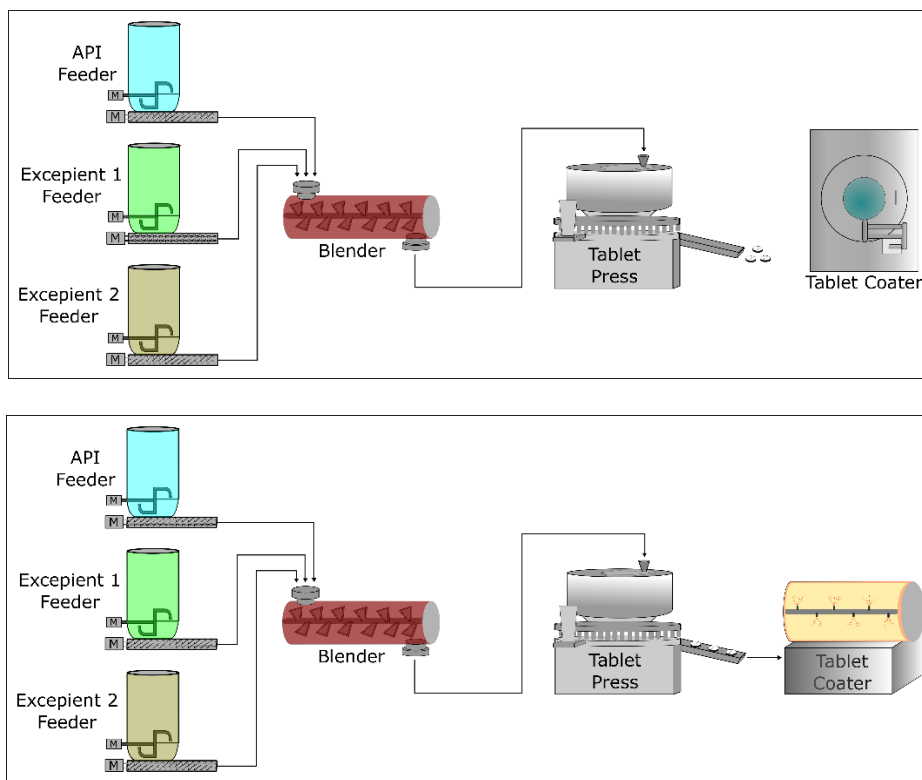


Figure 1: Schematic diagrams of continuous direct compression systems with a separated tableting coater operating in a batch mode (top) and integrated with a tablet coater operating

in a continuous mode (bottom) For a continuous capsule manufacturing process, the compression and coating step is replaced with an encapsulation step.

2. CONTROL STRATEGY

One unique aspect of the solid oral drug product CM is the continuous feeding of powdered materials. It can have a direct and significant impact on final product quality (e.g., assay and content uniformity) by introducing disturbances into the system. Therefore, the control strategy should be designed to minimize the occurrence of disturbances during the operation or if such disturbances do occur, mitigate their impact on the subsequent process steps (e.g., blending, granulation, and/or tableting), and ensure that the final product quality is not affected. The considerations presented below focus primarily on the control elements specific to address the product quality risk due to these disturbances. It should be noted that other risk factors for processing powdered materials (e.g., diversion of powder and granular mixtures that affect the content uniformity) still exist but may not be unique to CM and therefore are not discussed in this annex.

2.1. Material Characterisation and Control

Suitable risk analyses, experimental investigations, and/or modeling and simulation should be considered to evaluate the potential impact of material attributes and their lot-to-lot variability on material flow or process dynamics within the CM equipment (e.g., the loss-in-weight (LIW) feeders and blenders). These material attributes include, for example, the density, cohesion, particle size distribution, surface energy, powder rheology properties, and electrostatic propensity of the drug substance and excipients, and they should be characterised using appropriate methods. The understanding gained from these analyses and/or investigations during pharmaceutical development should be used to, for example, support the design of equipment and system configurations, determine the appropriate operating ranges for process parameters, and justify controls (specification) of input material attributes. Such understanding should be further enhanced throughout the lifecycle to identify and mitigate the impact of any potential changes in material attributes on the process performance and product quality.

For legacy solid oral products that are to be switched from batch to continuous manufacturing, the existing specifications for the drug substance and excipients should be reevaluated for its use in a CM system. These specifications may need to go beyond those used in batch manufacturing.

2.2. Equipment Design and System Integration

For feeding systems (e.g., LIW feeders), the flow rate and its variability should be characterised to demonstrate their accuracy and precision to meet the target formulation composition. It is important to characterise and understand the impact of the potential disturbances resulting from the feeding system on the downstream process steps such as blending and tableting. This understanding informs and supports what other control strategy element(s) will be needed to mitigate the impact of these variations on product quality.

The knowledge on the nature of potential disturbances (e.g., amplitude and duration) that may be generated from the feeding systems should be taken into consideration for the equipment design or selection of downstream processing steps. For example, the blender and other process steps or components of the system (e.g., channels transporting materials between process steps) could be designed or selected to provide a desired level of back mixing to mitigate the impact of these disturbances on the final product quality. Control of the relevant aspects of equipment and system design or configuration should be part of the control strategy.

For system integration, material flow should be coordinated across all process steps to avoid any material accumulation or emptying along the process. The mass balance across the integrated process steps requires a thorough understanding of material flow and RTDs of each process step within the intended operating ranges. This RTD information is important to understanding disturbance propagation and tracking material through transformations or process steps.

2.3. Process Monitoring and Controls

The CM of solid oral drug products generally exhibits relatively fast process dynamics (e.g., the residence time of blenders is on the order of a few minutes). For example, the LIW feeder may introduce disturbances in the mass flow rate that can be large in amplitude and short in duration (sometimes on the order of a few seconds). Fast dynamic responses may also occur during changes in operating conditions (e.g., changes in the flow rate or input material attributes). Therefore, monitoring and control of these disturbances or fast dynamics are key elements of the control strategy to ensure this type of CM systems remains in a state of control.

The sampling strategy needs to have a sufficient frequency to monitor the fast dynamics or detect rapid changes. Specifically, it is important to determine the maximum rate of change for the process and align the frequency of measurements with that rate, so all process changes outside the acceptable range will be detected.

Once monitoring of relevant parameters is established by using appropriate PAT tools (e.g., spectroscopic and chemometric tools as well as non-spectroscopic sources and soft sensors), appropriate control (e.g., alarm and action limits and active controls, see Section 3.1.5 for general considerations) should be incorporated to minimize the occurrence of disturbances in these parameters.

The real-time data on process parameters and quality attributes, together with the understanding of the system's RTD, provides high detectability of disturbances. Therefore, this information enables accurate material diversion, when the disturbances persist and propagate through the system and affect the quality of in-process materials.

2.4. Process Models

As described in the main ICH Q13 guideline, process models can be used to support the risk assessment and the design of control strategies. Process modeling can also be an integral part of the control strategy. Annex V provides examples to illustrate various applications and

corresponding considerations of the RTD model for the CM of solid oral drug products at the low, medium, and high impact levels.

3. CONTINUOUS PROCESS VERIFICATION

As described above, because of the need for using PAT tools to assess, monitor, and control transient disturbances and support other elements of the control strategy, the CM of solid oral drug products naturally generates a large volume of real-time data on the system performance and material quality during pharmaceutical development and commercial production. In addition, this type of CM system utilizes the same or highly similar equipment for pharmaceutical development and commercial production. These unique features make CPV well suited to the evaluation of the solid oral drug product CM by utilizing data from batches generated throughout development and commercial production to validate the CM system in a continuous basis.

The amount of information generated by the CPV approach is much larger than that from the traditional validation approach. As a result, for solid oral drug product CM systems, there is an opportunity that fewer batches are needed for process qualification. In addition, comprehensive product and process understanding, as well as information generated continuously from the real-time monitoring of critical process parameters and quality attributes could allow process qualification to rely on batch sizes less than the proposed maximum batch size in the dossier (e.g., shorter run times). Furthermore, CPV may support post approval changes without prior approval or with lower notification categories, such as changing the commercial production runtime outside the range approved in the dossier (e.g., extending the runtime beyond the approved maximum runtime) due to changes in the market demand.

ANNEX III: CONTINUOUS MANUFACTURING OF THERAPEUTIC PROTEINS

1. INTRODUCTION

This annex is intended to provide additional information and guidance specifically for therapeutic protein drug substance produced using partially or fully integrated CM methods. Manufacturing processes utilizing a single continuous process step, such as a perfusion bioreactor, in the absence of integration with an adjacent process step are out of scope. Aspects of CM not specifically addressed in the main guideline regarding control strategy and validation for therapeutic proteins are detailed in the subsequent sections. It is acknowledged that alternative strategies and approaches are possible.

2. BATCH SIZE

Similar to batch processes, a CM process drug substance batch may represent a single working cell bank (WCB) vial thaw, cell culture, and purification, resulting in a single drug substance batch. In some CM processes, a single WCB and cell culture can generate multiple drug substance batches. Where multiple harvests from one cell culture run are collected, it should be demonstrated that the increasing cell age during the culture run does not have an impact on the quality of the

drug substance. Intra-batch (i.e., derived from initial harvest through to last harvest) and inter-batch (i.e., derived from different cell culture cycles) data should demonstrate the consistency of drug substance quality.

3. CONTROL STRATEGY

3.1. Adventitious Agent Control

It is important for the applicant to demonstrate that appropriate control of potential adventitious agent contamination is in place based on the risk assessment of their CM equipment, system, and facility. In addition to current regulatory expectations for adventitious agent testing, a periodic risk assessment should consider the potential for unprocessed bulk to be continuously harvested from the bioreactor during processing, which is unique to CM. Therefore, a scientifically justified testing strategy tailored to the risk profile of the host cell lines and/or product, as well as the potential contamination from the use of animal-derived raw materials or other risks for unprocessed bulk harvest, should be developed. These strategies should include the type and frequency of adventitious agents testing undertaken to demonstrate the process is free of contamination during cell culture. Rapid testing for microbial and adventitious agents may support the implementation of CM approaches and real time decision making for mitigation responses.

3.2. Equipment Design and System Integration

Equipment suitability and qualification for CM processes should assess extractables and leachables from the equipment and the stability of the equipment for the maximum run time and/or cycles. The large-scale implementation of closed processing equipment may decrease the risk of the spread of adventitious agents from upstream to downstream processes. Special consideration should be taken to ensure the integrity of single-use equipment to prevent contamination risks. CM equipment, system integration, and facility designs (i.e., physical diversion of post viral inactivation material vs. ballroom layouts with a closed system) should consider the overall risk reduction strategy to minimize the potential for inadvertent contamination. In addition, for operations where surge vessels or tanks are utilized for flow-rate adjustments or other purposes between steps, the uniformity and microbial risks to the products in these surge vessels or tanks should be evaluated in advance.

3.3. Process Monitoring and Real Time Release Testing (RTRT)

While CM lends itself to a variety of monitoring schemes varying in their degrees of automation. Regardless of the approach, controls should be in place at appropriate stages of the CM process, to allow for timely data acquisition and analysis to ensure the state of control operation, and potentially adjust the relevant process parameters to ensure the quality of output materials. With increasing online and inline PAT capabilities and the development of automation systems for equipment monitoring, coordination, and process tracking, a continuous monitoring scheme may be implemented. Such an approach can be supportive for RTRT. However, for therapeutic proteins, a phased approach may support RTRT for some quality attributes, while other currently less amenable tests may necessitate that full product release is not initially in real time. For example, confirmatory sterility testing needs to be performed for each lot, even if there is online/inline rapid testing for microbial contamination until there is adequate assurance of method replacement.

4. PROCESS VALIDATION

4.1. Approach to Process Validation

The number of batches required to validate a process may vary according to process complexity. Validation activities for the upstream process should include a sufficient number of runs to demonstrate consistency of the entire cell culture process. As multiple harvests are typically generated under continuous production (e.g., perfusion bioreactor), multiple drug substance lots may be purified from a single WCB thaw. Additionally, downstream process validation activities should demonstrate the lack of impact on product quality attributes for the entire cell culture duration as well as inter-batch consistency.

Alternatives to the traditional process validation approach may be considered. For example, a CPV approach may be considered to support process validation and should be described and justified as part of the dossier (see Sections 3.3 and 4.7 for general considerations). In conjunction with CPV, a risk assessment and the applicability of the small-scale development data, process models, as well as understanding cell line variability may provide justification to support differences to the traditional process validation approach.

Unique considerations in process validation of CM are as follows:

- Validation/qualification of equipment and PAT/automation systems during worst-case (e.g., certain process parameters at high and low ends) and representative run times is important, especially in cases where these equipment and systems provide real-time monitoring and control of the process. Validation of sampling points and in-line, at-line, and/or on-line equipment calibration associated with process control or release testing should also be included as part of the dossier.
- Validation/qualification of a CM process should incorporate start-up, shutdown, and pauses (if applicable), evaluate lower and upper range of run times, and demonstrate the ability to detect disturbances which require diversion of non-conforming material.
- Validation/qualification of changes in production rates, process step linkages, surge vessels, etc. is also important to ensure a state of control operation. These validation/qualification studies should, for example, consider potential hold times for diverted material, potential re-introduction of diverted material back to the process, and potential process pauses arising from surge tank capacity due to changing process flow rates.

4.2. Run Length Considerations

Bioreactors for CM may be run for significantly longer periods of time than batch bioreactors. The requirements for the establishment of cells at the limit of *in vitro* cell age (LIVCA) and for product characterisation at the end of a production bioreactor run do not differ regardless of the mode in which the bioreactors operate. However, it should be emphasized that previously established LIVCA for a batch mode bioreactor run may not be applicable to a bioreactor run in a continuous mode as typically both are run under different conditions. Other factors to be considered should include, but not be limited to, the control of adventitious agents and the impact of prolonged use

on equipment performance. In all cases, experimental data and risk assessment tools should be used to determine the testing frequency of adventitious agents, as well as to ensure adequate equipment and process performance for the duration of the run.

4.3. Filtration

Filtration steps in CM may be subject to longer filtration periods and potentially increased throughput per unit area or a greater number of filter changes than those under batch manufacturing. Given these factors, a control strategy and a clearly defined scheme should be put in place to allow for filter changes and post-use integrity testing, as appropriate, without interrupting the process. In the event of a filter failure, a clear strategy for material diversion and refiltration (reprocessing) should be defined.

4.4. Viral Validation

The general requirements outlined in ICH Q5A for viral safety and clearance remain applicable for CM, including the qualification of small-scale models, which should consider aspects unique to a continuous process.

4.5. Diversion / Reintroduction

As in batch manufacturing, situations may occur which necessitate diversion of material that does not conform to pre-defined operating ranges for process parameter or acceptance criteria for quality attributes. Upon determining root cause, reintroduction or reprocessing may be conducted with validated processes, hold times, and under CGMP control. Scientific justification should be provided for the purpose and length of the pause, taking into consideration the overall process hold time (inclusive of anticipated pauses) and its impact on product quality attributes and microbiological control. Consideration should also be given to the traceability of any reintroduced material. Hold time studies should be conducted with special consideration taken for leachables and extractables of the single-use systems.

4.6. Raw Material Variability

A CM process may use several lots of raw materials throughout a single campaign due to the extended cell culture run times. A risk assessment of the raw material variability along with traceability of the raw material for product disposition in the event of an out-of-specification (OOS) test result in the process should be defined.

ANNEX IV: INTEGRATED DRUG SUBSTANCE AND DRUG RPRODUCT CONTINUOUS MANUFACTURING

1. INTRODUCTION

This annex provides additional clarification on CM-specific aspects that should be considered in the development and implementation of integrated drug substance and drug product CM processes (referred to as integrated processes hereafter). General considerations that apply to drug substance and drug product are not discussed here. A hypothetical process for small molecules is used to illustrate key points; other approaches could also be used.

2. INTEGRATED DS/DP PROCESSES

2.1. Key Characteristics of Drug Substance and Drug Product Process Steps

Table 1 shows several differences between drug substance and drug product processes. These differences should be considered in the design of an integrated process as the design impacts, for example, process operation, performance, and wastage amount. Some aspects to consider include the selection of equipment including equipment design, equipment connections (and surge lines when needed) to manage differences in residence time, location of material diversion and in-process measurement points considering the method and time taken for measurement, among others.

Table 1: A Comparison of Drug Substance and Drug Product Processes

Aspect	Drug Substance Process	Drug Product Process
Input Materials	Solvents, gases, starting materials, intermediates, catalysts, and others	Drug substance and excipients (solids and liquids)
Material Addition Methods	Pumps and mass flow meters (for liquid or gas addition)	Loss-in-weight feeders and pumps
RTD	Short RTD, typically seconds to minutes in reaction; Long RTDs in crystallisation and workup, typically hours	Short RTDs, typically minutes
In-Process Measurements and PAT	Lower frequency measurements for long RTD process steps; High resolution analytical methods for impurity detection	High frequency measurements for short RTD process steps
Response Time for Diversion	Typically seconds in reaction; Hours in crystallisation	Seconds
Typical Run Times	Days to weeks	Hours

2.2. Example of an Integrated Process

The drug substance is synthesized through a three-step chemical synthesis using flow reactors and purified through solvent extraction and continuous crystallisation to obtain a crystal slurry. Crossflow filtration is used to concentrate the crystal suspension, remove the mother liquor from the process stream, and exchange the solvent to finally obtain a highly concentrated crystal slurry (i.e., drug substance). In this example, the drug substance and drug product processes are integrated through this continuous filtration line. The drug product is manufactured by a continuous wet granulation process. Excipients are fed through loss-in-weight feeders and blended using a helical ribbon blender. The excipient blend is fed into a twin-screw extruder using loss-in-weight feeders. The concentrated drug substance from the filtration is fed into the extruder and functions as both the drug substance source and the granulation fluid. The resulting wet granules are dried in a continuous fluid bed dryer. Dried granules are blended with extra-granular excipients and compressed into tablets using a tableting machine. The compressed tablets are dedusted and film-coated using a continuous coating machine.

Figure 1 provides a flowchart for the integrated process example. The process flowchart shows the following elements:

- Material addition points, distinguishing between solids and liquids
- Individual process steps used for DS and DP manufacture
- Process design for the interface between drug substance and drug product
- All in-line/at-line/off-line measurements and the attributes measured
- Surge lines or tanks, if used
- All diversion points

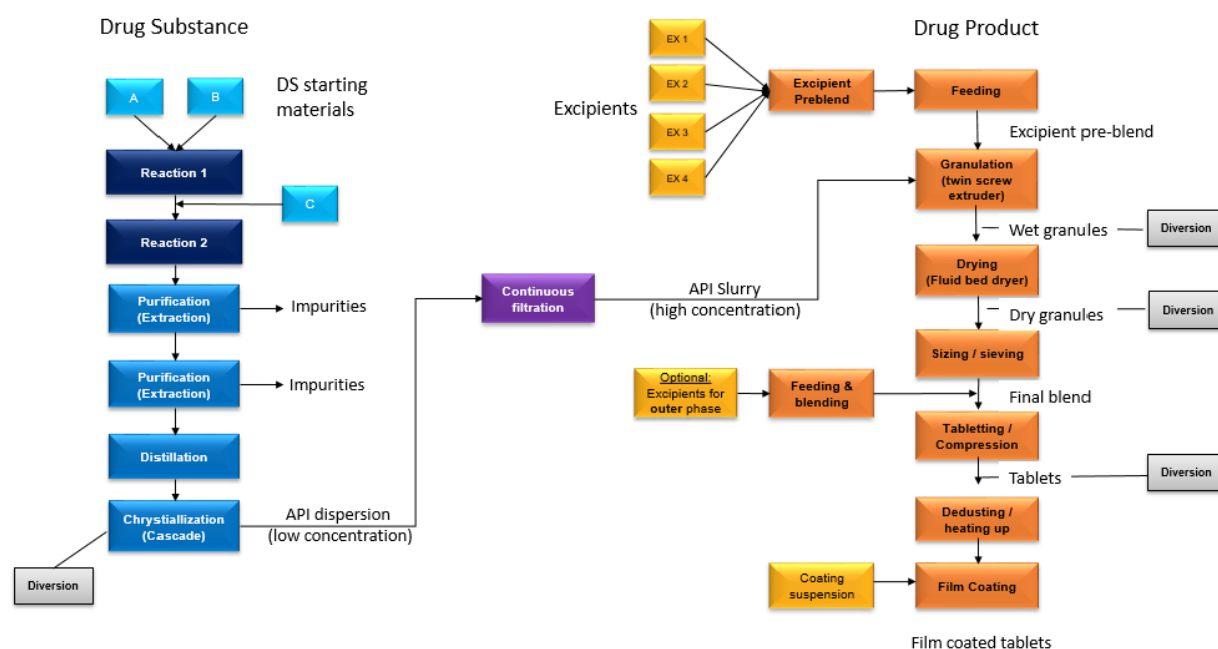


Figure 1: Example of an integrated drug substance and drug product CM process

2.3. General Considerations for an Integrated Drug Substance/Drug Product Process

Operational aspects, material traceability, in-process test methods, diversion strategy, and other factors influence the design of surge lines/tanks and the locations of measurement and diversion points. Surge tanks/lines help manage transient disturbances, accommodate slower at-line or off-line testing (e.g., chromatography methods) and balance different RTDs, but could blur material traceability and increase discard amounts when disturbances occur. In this example, individual equipment and connections to one another are designed, such that no surge lines or surge tanks are used to accommodate the differences in residence time.

Continuous crystallisers are used for impurity removal in this example as crystallisation is a common method of purification. Depending on the type of drug substance and impurity profile, other purification methods (e.g., extraction, distillation, and other separation technologies) that give the desired degree of purity may be used instead of crystallisation.

Process analytical instruments are located at the equipment exits prior to the diversion points for ease of diversion when out-of-specification readings are obtained. For the drug substance segment of this process, a single diversion point (#1) is located after crystallisation (prior to continuous filtration). The drug product segment has three diversion points at the end of granulation (#2), drying (#3), and tablet compression (#4) steps. This system design is incorporated into the process monitoring and material diversion strategy as discussed in Section 2.7.

2.4. Startup and Shutdown

Individual steps of the integrated process achieve a state of control at different times due to differences in the type of transformation (e.g., chemical and physical) and the residence time in

the equipment. For example, drug substance process steps typically take longer than downstream drug product process steps to reach a state of control. Careful planning of startup and shutdown sequences enables faster product collection and reduces wastage. Various approaches could be used for startup/shutdown. One such approach is step-wise start-up of the individual process steps (from first to last). Each process step is operated with product diverted to waste until a state of control is reached before starting up the next process step. A similar step-wise sequence could also be used for process shut-down.

Product that is formed under a state of control within each process step may be collected; it is not necessary for the entire integrated line to be in a state of control for product collection. In the step-wise startup scenario described above, product collection may begin when the last process step reaches a state of control. When a step-wise sequence (from first to last) is used for process shutdown, product collection stops when the last process step is no longer able to maintain a state of control. Although by that time the upstream process steps are already shut down, the material collected was formed when each process step was in a state of control.

Time taken for integrated processes to startup, reach a state of control, and/or shutdown could impact operational decisions. For example, when these times are long, system pauses may not be readily built into process operation as a routine event (e.g., to accommodate operator shift changes). Long startup times may favor a campaign approach to manufacture (as illustrated in Figure 2) instead of manufacturing single batches with separate startups/shutdowns. In the integrated process example, events such as fouling of flow reactor or crystalliser, wetting or clogging of dryer filters, may not be observed when a single batch is manufactured, but could occur over the longer run times associated with a campaign. When the process is operated as a campaign, the ability to operate continuously over the campaign duration (i.e., continuous run time) and maintain consistent product quality should be demonstrated.

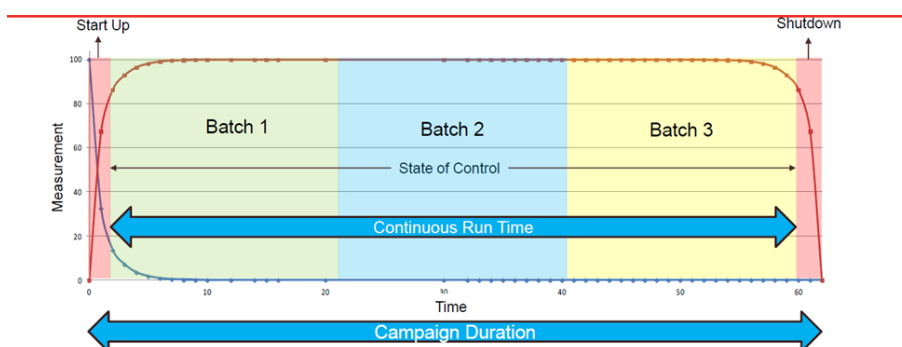


Figure 2: Operation of an Integrated Process as a Campaign

2.5. Process Monitoring and Control

A process design approach considering both individual and integrated steps should be used to develop an effective control strategy, as some quality variations in earlier steps due to disturbances or process variability may be resolved by downstream processes and may not require material diversion. Allowable variations (including minor disturbances) identified through design of

experiment or other suitable studies could be incorporated into the process control strategy. For example, in the annex process, the process parameter ranges for material additions and reactors could be based on the variations shown to be within the purification capability of the crystallisation step and have no adverse impact on the drug substance purity/impurity profile. During routine operation, measurement at location #1 by a suitable method (e.g., ultra-performance liquid chromatography) provides further assurance of product quality.

In this example, the location of the monitoring/diversion points creates process segments (e.g., the drug substance process steps up to monitoring location #1, process steps from location #1 to #2, and so on). The sampling and diversion strategy may be based on RTD characterisation of the entire process, process segments (as described above), or individual steps. Appropriately designed process segments could facilitate process monitoring and diversion without the use of surge lines or multiple diversion points (which may contribute to disturbances). The RTD of the drug substance process segment provides a suitable time frame to enable an appropriate sampling frequency (for adequate coverage of product quality in real time) while considering the time needed for measurement considering the method and instrument capability. The sampling and diversion strategies are based on a low-impact RTD model that incorporates potential variation in process dynamics (see Annex V).

Ongoing assessment of equipment performance helps predict and prevent potential problems and ensures the ability of a CM process to operate as intended over time. Two examples of equipment performance indicators for the process described above are: (i) during continuous filtration, monitoring filter back pressure to evaluate filter saturation (maximum pressure) and prevent filter failure, and (ii) during material feeding using loss-in-weight feeders, monitoring the feeder screw speed in relation to its maximum capacity to inform low feeder fill-level. Performance indicators such as these are not the primary process control parameters and are not direct constituents of the control strategy. However, an overall discussion on how process control will be ensured, especially during long running times, should be provided (see section 4.2. of the core guideline). CPV might be used in addition to continuously monitor manufacturing performance.

2.6. System Dynamics and Material Traceability

Refer to the main guideline and Annex V. The tracer used for RTD characterisation needs to be as representative as possible, behaving the same way as the commercial material with regards to system dynamics. The tracer is analyzed qualitatively and quantitatively at all possible observation points. The amount of tracer used in the studies depends on the detection limit for the tracer at the defined observation points. The amount of tracer used may affect its representativeness and hence need to be carefully selected.

2.7. Transient Disturbances and Diversion Strategies

Diversion limits should reflect process knowledge at diversion points. The attribute affected by the disturbance and the consequences to downstream operations could determine which diversion point to use. Each process should be evaluated on an individual basis. Disturbances are not always detected by PAT measurements; loss of the state of control may be detected through system indicators that signal a potential for material diversion. Disturbance limits for system indicators

are set considering the entire process, since some disturbances may be resolved in subsequent operations.

Some general strategies for handling transient disturbances are given below.

- When the disturbance cannot be tolerated, material may be diverted at the next diversion point. This is usually for severe disturbances where material should not be allowed to proceed through the process. Diversion of material within the line requires a compensation strategy for the missing material in flow such as reducing throughput at the remaining down-stream line. However, the impacts of changes in mass flow need to be considered (e.g., for material traceability and RTDs).
- When a disturbance is detected, surge tanks or lines can be used to temporarily collect material. Surge tanks/lines help manage transient disturbances, accommodate slower at-line or off-line testing (e.g., chromatography methods) and balance different RTDs. Buffers intended to collect material such as surge tanks and lines may compromise material traceability and increase discard amounts if quality attribute out-of-specification results are obtained.
- For disturbances that are expected to resolve in subsequent operations, the process can continue running up to the last possible diversion. RTD needs to account for both spread and mass flow. Diversion, if still necessary, is then carried out at the last diversion point to ensure continuous material mass flow and to prevent disruption or mass flow compensation in the downstream operations.

In case of a disturbance or process event where the predefined acceptance criteria are no longer met, the potentially impacted material should be “flagged” and lead to a material diversion point of product to waste.

The amount of material diverted is based on state of control and how it is linked to product quality. The sampling strategy and process dynamics, as well as the ability to manage, trace, and remove materials are considered in establishing the thresholds for diversion. Additionally, a risk-based safety margin is included in the established thresholds to ensure all non-conforming material is diverted from the batch.

3. SPECIFICATION AND BATCH DATA

3.1. Drug Substance Specification

Institution of a drug substance specification facilitates the management of lifecycle activities, such as bridging manufacturing or facility changes, investigating adverse events and product recalls, and facilitating the development of pharmacopeia standards. Even though the drug substance is not isolated in an integrated drug substance and drug product process, a drug substance specification should be defined and follow the recommendations in the relevant ICH guidelines. Although a drug substance specification is instituted, drug substance testing is not required on a routine basis when the integrated process is appropriately controlled. A set of process performance

criteria should be defined such that the drug substance could be considered “conforms to specification, if tested” when the process performance criteria are met. Additionally, conformance to the drug substance specification should be verified on a periodic and event-driven basis by testing the purified drug substance at an appropriate location using a relevant sampling plan. The sampling location should be incorporated into the process design to enable this evaluation (e.g., diversion point #1 in this annex example). The frequency of the periodic verification should be defined and can be based on the frequency of production (e.g., every 10th batch) and time (e.g., annually), whichever occurs first. Event-based verifications could be triggered by a change in supplier, starting material, synthesis conditions, or other factors. Similar to the ICH Q6A guidelines for periodic testing, when tested, any failure to meet the drug substance specification should be handled by proper notification to the appropriate regulatory authority(ies). If these data demonstrate a need to restore routine testing, then each batch should be tested.

3.2. Drug Product Specification

When the drug substance and drug product processes are integrated, attributes typically associated with the drug substance quality should be included in the drug product specification when appropriate. Therefore, the drug product specification in an integrated drug substance and drug product process is more extensive compared with that of a batch process. The drug product specification could include test and acceptance criteria for related substances, residual solvents (used in drug substance synthesis), heavy metals, drug substance process impurities, and/or polymorphism, among others. Sampling location should be appropriately identified in the drug product specification as some testing may need to be performed following purification. Table 2 shows potential test attributes that could be included in the specification, when applicable, for a new drug product and drug substance made by an integrated process.

Other approaches (e.g., real-time release testing) could also be used, when appropriate. Quality attributes determined through real time release testing using PAT do not need routine testing at the end drug product stage but can be included in the finished product specification and noted as “comply, if tested.” Additional elements of the control strategy are considered for decisions regarding product release.

Table 2: An Approach to Drug Product* and Drug Substance* Specification Testing
(for the integrated process example in this Annex)

Test Attribute (when applicable) ¹	Drug Product Specification (Every batch tested)		Drug Substance Specification (Periodic testing)	
	Test	Sampling Location ²	Test	Sampling Location ²
Description	✓	Coated tablets	N/A	N/A
Identity	✓	In-process/PAT or tablets	✓	In-process/PAT or tablets. May use DP test result
Crystalline Form	N/A	Not tested (with lack of form change is demonstrated)	✓	Case Specific. Sampling Point S1
Chirality	N/A	Not tested	✓	Diversion Point 1
Particle Size	N/A	Not tested	✓	Diversion Point 1
Assay	✓	Core tablets or in-process PAT/RTRT	N/A	
Impurities ³		Diversion Point 1 (at-line HPLC) and Coated Tablets (off-line testing), as appropriate		Diversion Point 1 (at-line HPLC)
Related Substance	✓		✓	
Residual Solvents	✓		✓	
Elemental Impurities	✓		✓	
Genotoxic Impurities	✓		✓	
Dissolution	✓	Coated Tablets or Dissolution Model	N/A	N/A
Hardness / friability	✓	In-process testing during compression	N/A	N/A
Uniformity of dosage units	✓	Uncoated Tablets	N/A	N/A
Water content	✓	Coated Tablets (LOD)	N/A	N/A
Microbial limits	✓	Coated Tablets	N/A	N/A

¹ Includes tests that are necessary to ensure the identity, strength, quality and purity of the drug substance and bioavailability of the drug product.

² Tests that are common to both drug substance and drug product specification need to be tested only once; same test result can be used for the drug substance and drug product.

³ Impurity specification for drug substance and drug product may be different.

3.3. Batch Data

Batch data for both drug substance and drug product should be provided from representative commercial batches (e.g., made using the commercial integrated CM system, process and control strategy). When a range of batch sizes or run times are proposed in the dossier, the data could be

obtained from batches representing the run time at initial product launch (not necessarily the maximum run time proposed in the dossier).

Studies to support the robustness of the drug substance impurity clearance should be performed in the CM mode and represent the process dynamics proposed for the commercial process (e.g., input flow rates, integrated CM equipment, and RTD); some development studies could be conducted in batch mode or using a single process step. A discussion of potential impurities arising from the chemical synthesis, their fate in subsequent steps, robustness of impurity clearance and impurity carryover from drug substance to drug product (e.g., related substances, residual solvents, and catalysts) should be provided in the dossier. Refer to the ICH Q11 guidance. The control of impurities formation and clearance should be integrated into the overall control strategy.

4. STABILITY REQUIREMENTS

4.1. Drug Substance Stability

Drug substance stability data to define a re-test period is not applicable as the drug substance is not isolated and stored in an integrated process. However, institution of a hold time enables temporary storage of drug substance during a temporary interruption in production. Data to support the hold time may be obtained during development. In the absence of data to support a hold time, drug substance formed during a process interruption should be discarded.

4.2. Drug Product Stability

The ICH stability guidelines are generally applicable. However, some elements such as the selection of batches and definition of pilot scale (at least 1/10 of commercial scale or at least 100,000 units) for primary stability batches may not be fully compatible with continuous processes.

Consequently, in CM processes, samples collected from smaller batches, (e.g., defined by shorter run time) could be used in the primary stability studies as long similar processing conditions are representative of the commercial process. The use of these samples as primary stability materials is supported since they represent the manufacturing process which will be used for commercial manufacture. Product used for primary stability studies would be produced in a state of control and would be representative in terms of in process and intermediate controls and product quality to that manufactured for commercial supply.

5. LOCATION OF DRUG SUBSTANCE INFORMATION IN CTD

Although the drug substance is not isolated in an integrated process, information related to the drug substance and drug product should be provided in the respective S and P sections of the dossier as outlined by ICH M4Q. Placement of the process step that integrates the drug substance and drug product could be based on its relevancy to the respective section. For example, in the process provided in the Annex, the continuous filtration process could be included in the S section as it is related to concentration of the drug substance. Information that is not applicable to

integrated processes (e.g., drug substance container-closure and stability) should be noted as such in the dossier.

ANNEX V: PROCESS MODELS

1. INTRODUCTION

Process models, regardless of their intended purpose, are iterative and may change due to new knowledge, different context of use, new ranges or variability of inputs, or as part of a continuous improvement plan. This description holds true for empirical (data driven), mechanistic (based on knowledge of the system components), or hybrid (include empirical and mechanistic elements) models. Therefore, models are not static and are part of the lifecycle management. Figure 1 depicts the typical stages of a process model lifecycle.

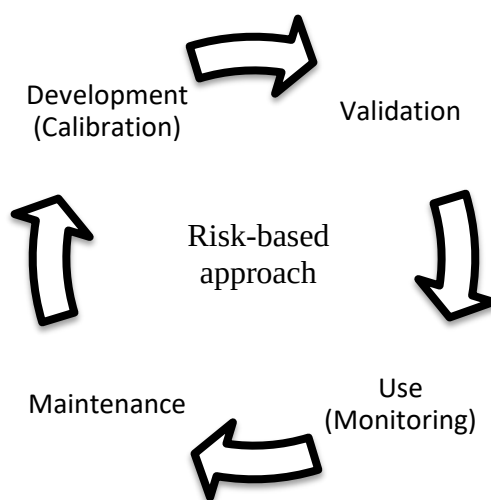


Figure 1: Typical high-level lifecycle of a model

Maintenance activities may trigger the need for an update to the model that can result in re-development and subsequent re-validation. The extent of activities associated with lifecycle stages should be commensurate with the impact level of the process model application.

This annex describes an example of the same model used in various applications with different corresponding impact levels. The lifecycle elements, as per Figure 1, are explored considering the three possible applications (i.e., use of process models to support sampling strategy, material diversion strategy, and in-process control), thereby resulting in three different impact levels (see case studies below). While the example refers to a continuous process for the drug product of a chemical entity, the same principles may apply for process models used for therapeutic proteins and drug substances for chemical entities.

2. EXAMPLE – A RESIDENCE TIME DISTRIBUTION MODEL FOR A CM PROCESS

This section illustrates the development of an RTD model in a CM process and describes how model development, validation, and lifecycle activities change based on the purpose and impact of the model. This is done through case studies based on the continuous direct compression process/system described below.

2.1. Process Description

A company manufactures tablets using the continuous direct compression process shown in Figure 2. The tablet has a high API/drug load. The process begins with feeding raw powders of the drug substance and excipients as input materials using loss-in-weight (LIW) feeders into the continuous blender. Material exiting the blender flows through a transport pipe to the tablet press where it is compacted into tablets. The near-infrared (NIR) probe in the tablet press feed-frame monitors API potency. The port located at the end of the press permits acceptance or rejection of tablets.

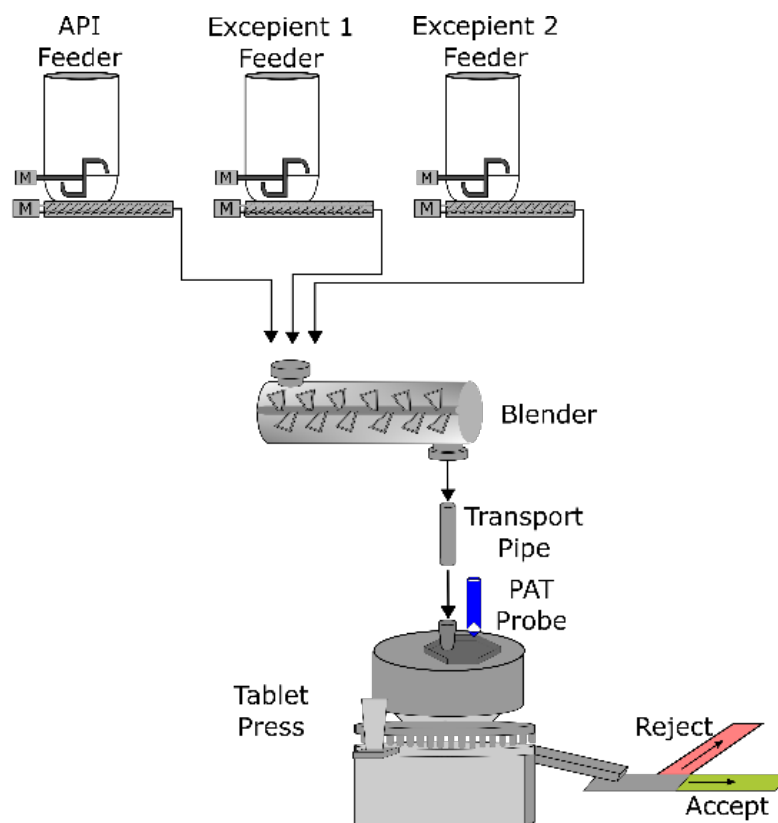


Figure 2: Schematic of a continuous direct compression process

Material attributes are controlled within pre-defined ranges. The process parameters¹ controlled here are material feed rates, blender speed (revolutions per minute (rpm)), blender holdup mass,

¹ The process parameters listed above are for exemplification purposes; additional process parameters may apply.

transport pipe fill-level, tablet press turret speed, compression force, and fill depth of die. These process parameters are operated at pre-defined set points within established high/low ranges.

2.2. Model Background

The objectives of the model need to be defined prior to developing the RTD model; then a mathematical approach is selected. RTD models are hybrid models that combine mechanistic understanding and empirically estimated model parameters. In this example, an RTD model is developed to represent the mixing (dispersion) and the transit of material (residence time) for the process and process conditions described above. Different mathematical approaches can be used to develop the RTD model including tanks-in-series and convective-diffusion models. The selected mathematical approach depends upon the process (materials, equipment, and operating ranges) being modeled and the context of use.

In an RTD model, model inputs are generally real-time measurements of process parameters and material attributes and the model outputs are predictions of the downstream concentrations of the formulation components. In this example, the RTD model inputs are real-time measurements of the input material feeding rates from the LIW feeders (drug substance mass flow rate, in this case) and blender speed while the model output is the API concentration prediction at the tablet press feed-frame.

The process dynamics are dependent on the actual process conditions. For example, higher material feed rates, higher blender speed, and lower transport line levels (within the defined ranges) can result in faster process dynamics, and vice versa. Material attributes that affect flowability (e.g., particle size distribution, density) are controlled within defined ranges such that process dynamics are not impacted. Other process parameters that can impact dynamics are controlled at set points (e.g., turret speed).

The following assumptions are used to develop the RTD model: blend flows uniformly without diversion; all individual excipients flow similar to the blend; and the method used to measure the process dynamics (see next paragraph) does not impact the process dynamics.

The RTD model contains model parameters that need to be estimated from experimental data. The model parameters are estimated by introducing a pulse or step change of a tracer into the process and measuring the concentration of the tracer as a function of time at a downstream location. Care should be taken that the method used to measure the process dynamics (e.g., flow characteristics and material attributes of the tracer, as well as size of pulse or step change) does not impact the process dynamics. The model parameters that need to be estimated will depend upon the mathematical approach. Typical examples of model parameters used in RTD models are mean residence time, delay time, number of tanks in series, and dispersion coefficient. The location(s) where the tracer concentration is measured will coincide with the location of the most accurate predicted concentration from the RTD model, which is the tablet press feed-frame, in this example.

The dynamic process response in these tracer experiments depends on the equipment design, process conditions, and the physicochemical properties of the formulation. The tracer experiments can be conducted at different process parameter setpoints, and RTD model parameters can be

obtained for each experiment. Empirical relationships can then be developed between process parameters (e.g., mass flow rate) and model parameters (e.g., mean residence time), allowing the model to incorporate real time changes in process parameters into the model predictions of API concentration.

Figure 3 shows the RTD variation over the range of process parameters that impact process dynamics in this process example (i.e., mass flow rate and blender speed) from a tracer pulse study. High mass flow rate and high blending speed represent the fastest process dynamics and low mass flow rate and low blending speed represent the slowest process dynamics. At the fastest dynamics, the tracer concentration at the feed-frame is negligible after ~220 seconds, while > 800 seconds are required to clear the tracer at the slowest dynamics. Hence a rapid disturbance (e.g., narrow pulses) is less spread out (i.e., narrower RTD) under faster dynamics compared to slower dynamics.

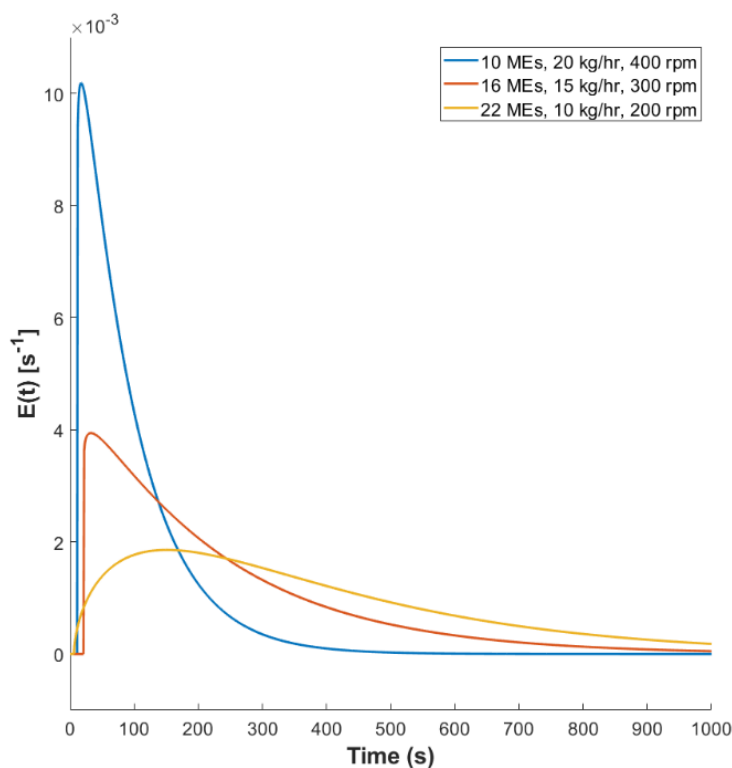


Figure 3: Measured RTDs as a function of the mass flow rate and blending speed ME represents the number of mixing elements (due to differences in the blender blade design) in the blender. The shear in the blender increases as the number of mixing elements increases from 10 to 22.

The measured tracer concentration over time can vary when experiments are repeated at the same conditions (i.e., process parameters). This variation is typically less than the variation observed when operating at different process parameters. The variability in the measured RTD at a given set of process conditions can be characterised through repeated experiments at those process

conditions. This variability can be compared to the variability observed over the proposed process parameter ranges. The quality of experimental data (e.g., number of points and precision of the measurement) impacts the confidence interval of the model parameters obtained from the parameter estimation. These confidence intervals are factored into the estimates of the slowest and fastest process dynamics.

The RTD for the process is dynamic. For example, the powder level in the transport pipe from the blender to tablet compression acts as a buffer; if the level increases the process dynamics will slow down, and if the level decreases the process dynamics will speed up. These dynamic responses are factored into the RTD model depending upon the context of use. Upon completion of model development, the model is validated. Model validation considerations are addressed under the case studies presented below.

The validated RTD model can be used to predict funnel plots that are specific to the formulation and process conditions used in model development, as shown in Figure 4. These plots indicate the downstream effect of a disturbance in material feed of a given amplitude and duration entering the system. Figure 4 shows that a disturbance of $\pm 20\%$ would not cause the API concentration to exceed the 90-110% limits until the duration of the disturbance exceeds approximately 220 seconds.

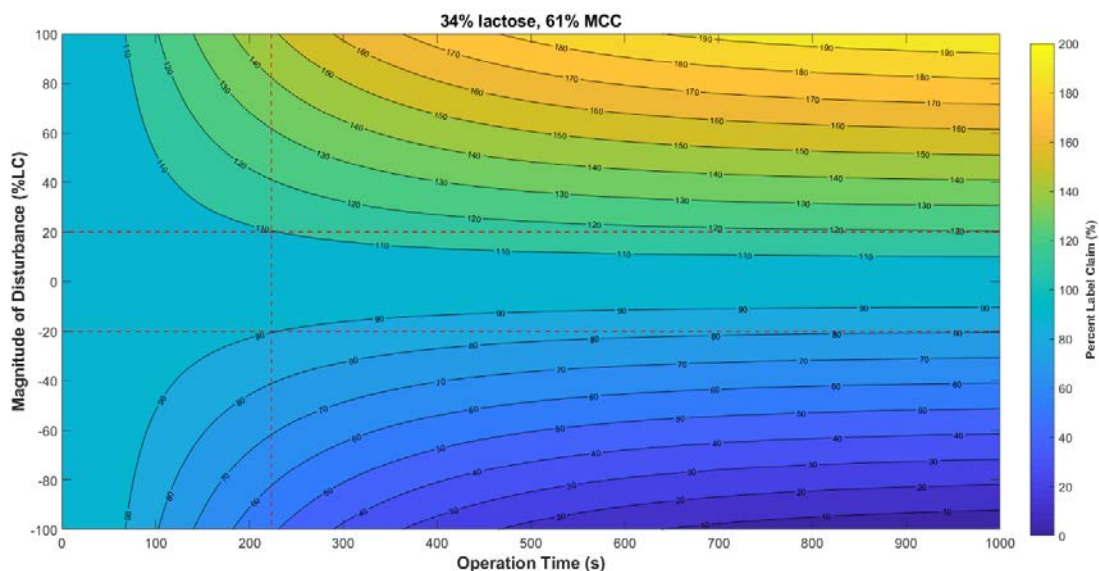


Figure 4: Funnel plots generated from the RTD model

2.3. Case Studies

2.3.1. Low Impact RTD model

Context of Use: The RTD model is used during development to decide on a sampling strategy. The RTD model provides information on how transient disturbances propagate and spread through the process and the amplitude and duration of disturbances that can impact product quality. A risk assessment can be used to identify likely ranges for the feeding disturbance amplitudes. The funnel plot can then be used to determine the maximum duration before these disturbances impact product quality. The sampling strategy should be designed to detect the identified transient disturbances

before the disturbance exceeds this maximum duration. As illustrated in Figure 3, Section 2.2, a disturbance may not be detected if the sampling frequency exceeds the RTD. Therefore, the sampling frequency of a CM process should be based on the fastest process dynamics expected within the defined process parameter ranges.

Impact Categorisation: During commercial production, blend potency is monitored by NIR. The model is used during pharmaceutical development to help develop a suitable sampling strategy. Therefore, the context of use for the model is Low Impact.

Model Development: The process parameters within the proposed operating ranges that result in the fastest process dynamics are identified based on development data and process understanding. The RTD for the process is measured at the identified process parameters using tracer experiments (i.e., a tracer pulse or a concentration step). Model parameters are estimated by fitting a tank-in-series model to tracer concentration data. This approach is justified. A risk assessment is performed, to define the conditions under which model revalidation occurs.

Model Validation: The agreement between model predictions of the tracer concentration over time and the experiments at the identified process parameters that represent the fastest process dynamics are evaluated and verified to meet the pre-defined acceptance criteria. Considering the model's context of use and low impact, the acceptance criteria can include qualitative visual comparison or quantitative root mean squared error.

Lifecycle Considerations: To keep the model current, adequacy of the model is re-assessed upon major process changes (e.g., new site or change in type of continuous blender) or process conditions change to a new region different or outside of the established design space. The reassessment should use the same acceptance criteria applied during model validation.

2.3.2. *Medium Impact RTD model*

Context of Use: The RTD model is used to develop in process control limits for material feeding and to trigger diversions related to material feeding. The feeder control limits are based on the funnel plot calculations as discussed under model development. Figure 5 shows how the model is used to trigger diversion of non-conforming material. If the model predicts a disturbance will cause the drug concentration (at the feed-frame) to be outside of the pre-defined acceptable limits, a signal is transmitted to the control system to reject tablets. Similarly, tablet collection is resumed based on the model prediction. To minimize risk, the calculation (programmed into the control system) of when to start tablet rejection is based on the RTD of the fastest observed process dynamic conditions, and the calculation of when to resume tablet collection is based on the slowest observed process dynamics conditions. In this example, the NIR sensor measures the API concentration at the feed-frame and is used to detect nonconforming material resulting from disturbances downstream from feeding. The utility of the model is that it can predict the impact of the process disturbance before it reaches the PAT sensor.

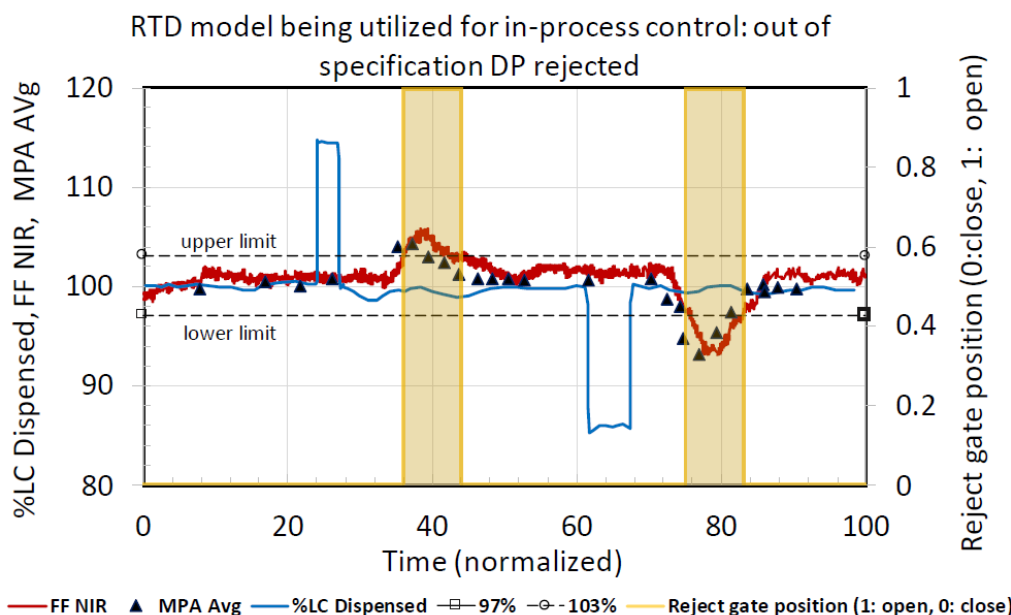


Figure 5: Schematic of RTD model-based diversion of non-conforming material

Impact Categorisation: The RTD model is identified as a medium impact model as it is coupled with a secondary measurement of API concentration (in-line NIR) downstream and therefore is not the sole indicator of in-process quality.

Model Development: The process parameters within the proposed operating ranges that result in the fastest and slowest process dynamics are identified based on development data and process understanding, and the respective RTDs are measured using tracer experiments as discussed previously. Model parameters are estimated by fitting a tank-in-series model to tracer concentration data. Two sets of model parameters are estimated: one set at the process conditions that represent the fastest process dynamics (e.g., high mass flowrate and high blender speed) and one set at the process conditions that represent the slowest process dynamics (e.g., low mass flow rate and low blender speed) for the proposed manufacturing process. A funnel plot is calculated using the RTD model that represents the fastest process dynamics (i.e., low amount of dampening). The funnel plot is used to propose material feeding limits. The RTD model representing the fastest process dynamics is used to calculate when to start diverting tablets if the material feeding limits are exceeded (i.e., when will the impacted material reach the diversion point). The RTD model representing the slowest process dynamics is used to calculate when to resume production collection once material feeding has returned to a state of control.

Model Validation: The agreement between model predictions of the tracer concentration over time and the experiments at the identified process parameters that represent the fastest process dynamics and slowest process dynamics are evaluated and verified to meet the pre-defined acceptance criteria. Tracer concentration is measured using an inline measurement for rapid sampling increasing the quantity of data available for statistical comparison. Experimental data used for model validation should be independent of the data used for model parameter estimation. Acceptance criteria for the model should use quantitative measures such as root mean squared error.

Lifecycle Considerations: To keep the model current, model performance is verified on a periodic basis and due to event-based triggers. Selection of the frequency of periodic reviews can be supported by an understanding of the risks that could impact the process dynamics over time. Lifecycle verification of model performance consists of tracer or spike experiments to verify that the slowest/fastest observed responses are still those identified at the time of filing and that the model acceptance criteria are met. If statistical analysis of the model prediction and experiments do not meet the pre-determined acceptance criteria, the RTD model parameters are re-estimated and revalidated. Event-based triggers for model verification include, for example, changes to equipment, process conditions different from the conditions used to validate the model, change in assumptions. When differences occur between the RTD model and the NIR model (secondary quality indicator) predictions, both the RTD model and the NIR measurement/model are re-assessed, redeveloped, and revalidated, as appropriate. A pre-determined statistical threshold is established to trigger such assessments of the RTD model and the NIR.

2.3.3. *High Impact RTD model*

Context of Use: RTD model used as in-process control for API concentration, without the NIR (i.e., no downstream secondary quality indicator). Model predicts API concentration at an appropriate location based on real time material feed rates and real time measurements of other process parameters that impact process dynamics (e.g. mass flow rate, blender speed, level of material in transport pipe). Material predicted to have an API concentration outside of the control limits is diverted. Release testing methods for assay and content uniformity are performed to confirm product quality using appropriate sampling plans. The usage of the RTD model in this context may be appropriate with adequate validation and when there is evidence showing the downstream disturbances can be mitigated by other element(s) of the control strategy (e.g., through equipment design and appropriate process parameter controls).

Model Development: A mechanistic understanding of the process is used to select the appropriate modeling approach, for example, due to low axial dispersion ($PE < 1$). A tank-in-series model is selected as the model form rather than an open boundary condition solution to the convective diffusion equation. Experimental development studies are conducted to verify all model assumptions, such as the absence of diversion downstream of the feeders. Studies are conducted to verify the method for measuring the RTD does not impact the process dynamics of the blend (e.g., comparing the material properties that impact flowability for the tracer and the blend, and conducting a tracer experiments at different pulse or step sizes). Model parameters are estimated over the range of process conditions and material attributes proposed for commercial manufacturing. This includes expected normal target process parameters as well as the maximum and minimum values for the proposed process parameter ranges. The RTD for the process is measured at over the proposed range of process parameters using tracer experiments (i.e., a tracer pulse or a concentration step). Model parameters are estimated by fitting a tank-in-series model to tracer concentration data. A mathematical relationship is established between process parameters and model parameters so the models can incorporate process variation into online predictions. Studies are conducted to evaluate if the model predictions are sensitive to variation in material properties for the given formulation. Experiments are conducted to demonstrate that the model parameters are not sensitive to variation in the ratio of formulation components.

Model Validation: The agreement between model predictions of API concentration and independent experimental data sets (i.e., data from experiments that were not used to estimate model parameters) are established per the pre-defined statistical criteria. Tracer concentration is measured using an inline measurement for rapid sampling, increasing the quantity of data available for statistical comparison. If the pre-determined acceptance criteria are not met, then the RTD model parameters should be re-estimated and validated. The model is validated over the intended processing ranges and include both the extremes of the proposed operating range as well as the nominal process conditions intended for commercial launch. Acceptance criteria should use quantitative measures such as root mean squared error. Statistical comparison between model predictions and experimental data should account for the range of model predictions based on the model parameter confidence intervals and experimental uncertainty.

Lifecycle Considerations: Verification of model performance is based on periodic and event-based triggers (e.g., equipment changes or if process conditions change to a new region different from the conditions used to validate the model). The frequency of periodic verification may depend on the number of batches produced annually, but it should be more frequent than an annual verification. Verification of model performance includes either tracer or pulse experiments to verify that the slowest and fastest observed responses are still those identified at the time of filing and that the model acceptance criteria are met. If statistical analysis of the model prediction and experiments do not meet the pre-determined acceptance criteria, then the RTD model parameters are re-estimated and re-validated. In the absence of a downstream secondary quality indicator, the forecasts from the RTD model can be contrasted with an alternative method for API concentration (e.g., at-line NIR, offline HPLC) using an appropriate sampling plan.

3. POST-APPROVAL CHANGE OF MODEL CONTEXT OF USE AND IMPACT LEVEL

An applicant may wish to change a model's context of use and, therefore, impact level post-approval. Such a change could be possible, if properly managed. Many aspects of the model, such as model type, development, validation, maintenance, lifecycle management, model accuracy, acceptance criteria, and amount of information submitted in the dossier, are dependent upon the model's context of use and impact level. A change from low or medium impact to high impact affects the above elements. Other aspects of the control strategy could also be impacted. For example, in the case studies presented in this annex, the measurement of the assay/content uniformity of the product and the statistical relevancy of the sampling plan could be affected when the model purpose and impact change from medium to high. A careful gap analysis of the requirements for low/medium and high impact models help identify the additional studies needed to address potential gaps and the information to provide in the dossier. Such a change should not rely solely on acceptable performance of the original model. Prior discussion and agreement with regulatory authorities on the bridging strategy, proposed data package, and implementation plan could facilitate a change in a model's context of use post-approval.

ANNEX VI: PERSPECTIVES ON MANAGING TRANSIENT DISTURBANCES

1. INTRODUCTION

A CM process and its control strategy are designed to maintain a state of control operation and minimize the risk of producing non-conforming materials. However, there could be transient disturbances over the entire production run, where it becomes necessary to isolate a portion of a batch of the material affected by these disturbances from the main portion of materials produced in the same run. This annex describes the risk-based approaches to manage different types of disturbances that may occur during CM operation.

Disturbances generally can be categorized into three types: (1) Type 1 where a single disturbance is observed, and its amplitude and duration are within predefined acceptance criteria; (2) Type 2 where a single disturbance is observed but its amplitude and/or duration exceeds the predefined acceptance criteria; and (3) Type 3 where multiple Type 1 disturbances are observed in the same CM run. As illustrated in the decision tree (Figure 1), each of these disturbance types exhibits a different level of risk to a state of control operation, and therefore warrants a different approach to evaluate and mitigate the impact of disturbances on the quality of output material. Case studies of a continuous direct compression system are provided below to detail the risk-based approaches for managing these three types of disturbances.

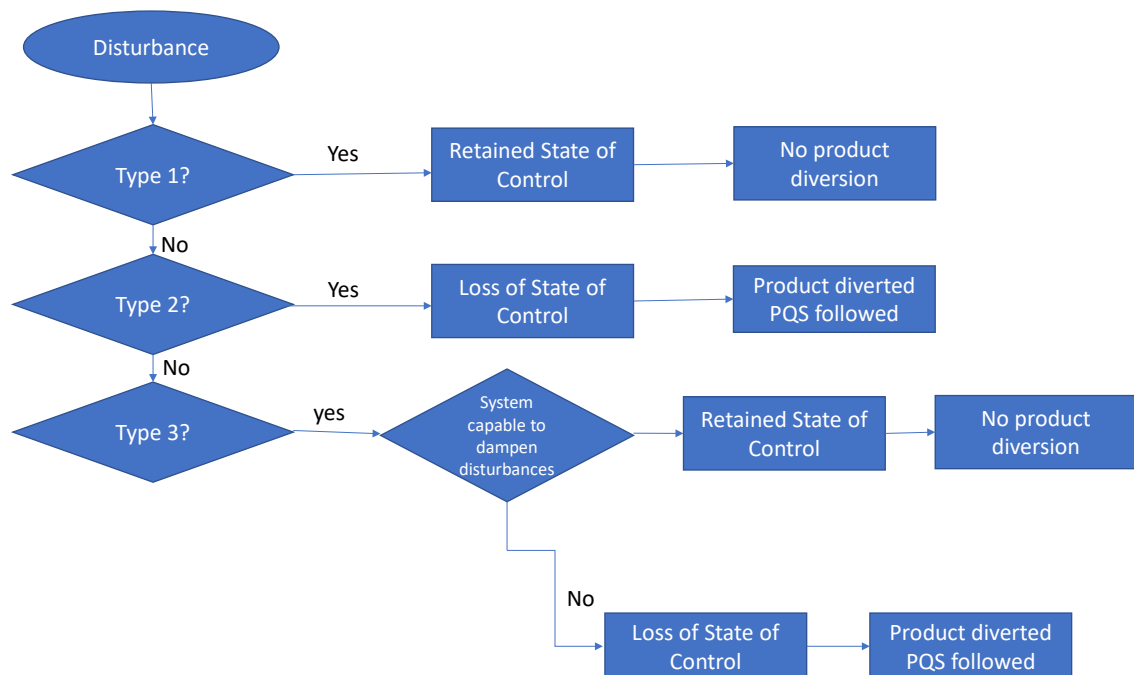


Figure 1: A high-level decision tree for risk management of three different types of disturbances that may occur during CM operation

2. BACKGROUND

A company has a continuous direct compression system composed of a set of LIW feeders integrated with a continuous blender and tableting machine (Figure 2).

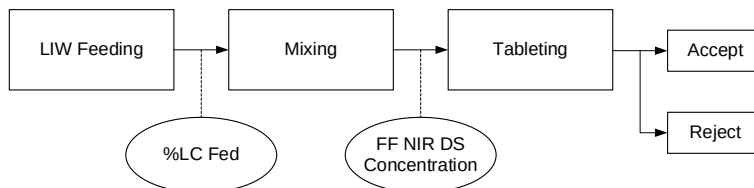


Figure 2: A schematic diagram of a continuous direct compression system

Multiple different equipment configurations were studied. The company optimized the equipment design and system integration in order to minimize risk of non-feeder related disturbances. A process model was developed to understand and characterise the system by accounting for, among other things, the mixing in the direction of flow (dispersion) and the transit of material (residence time). This process model predicts the drug substance concentration in the tablet press feed frame using the information on the quantity of the drug substance being added from the LIW feeders into the system. However, the decisions to collect or divert product during CM operation is done through the near-infrared (NIR) spectroscopy as a PAT tool to continuously measure the drug substance concentration in the tablet press feed frame in real time.

The company identified multiple key aspects that need to be considered for a state of control operation of its CM system. These aspects include, for example, raw material control, process monitoring (including disturbance detection), product collection criteria, integrated process controls, real time handling of disturbances, material traceability, and others as determined by the company based on the control strategy and system understanding.

Based on the above considerations, the company developed a control strategy and defined seven elements that need to be within the predefined criteria to allow material collection (Figure 3). Depending on the nature of disturbance (i.e., amplitude and duration), the process may proceed without interruption or diversion of materials, continue with appropriate diversion of tablets, or pause or stop to allow investigation into and if needed, remediate the root cause for the observed disturbance.



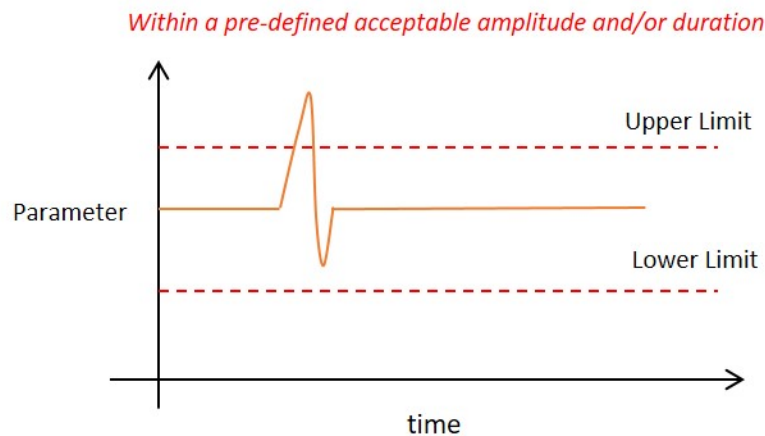
Figure 3: A schematic diagram to illustrate the relationship of control strategy elements to material collection and diversion.

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1132 3. CASE STUDIES

1133 3.1. Case #1

1134 **Type 1 Disturbances:** A transient disturbance which does not exceed the pre-defined amplitude
1135 and/or duration (Figure 4) beyond which it could impact the output material quality.
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1138 **Figure 4: A schematic representation of a Type 1 disturbance**
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1140 **Example:** A LIW feeder disturbance with amplitude and/or duration values that do not require
1141 material diversion.

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1143 **Possible Scenario:** One of the feeders encounters flow disruption causing a transient increase in
1144 the drug substance concentration (100 to 115% LC for 40 seconds) from the feeder to the mixer
1145 before returning to 100% (see Example 1 in Figure 5). Nevertheless, all elements are within the
1146 pre-defined criteria.
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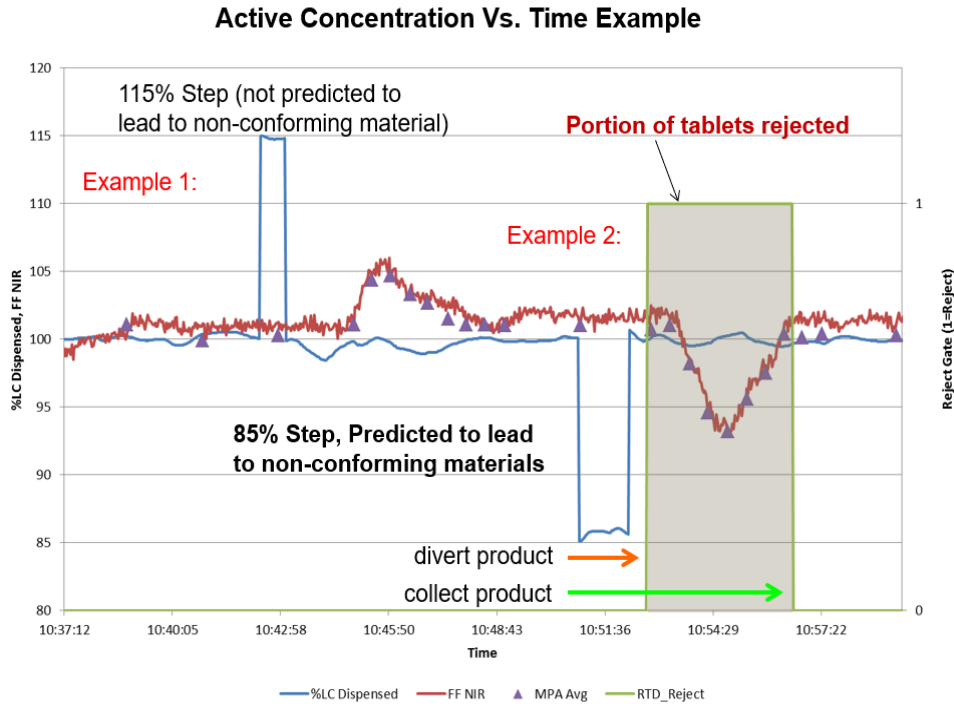


Figure 5: A diagram of drug substance concentration vs. time for two examples of transient disturbances

Impact on Quality: The process model predicts the drug substance concentration within the press feed frame will remain within the pre-defined acceptance limits of $100 \pm 7\%$, due to the back mixing in the blender. The NIR measurements confirm this prediction.

Response: Though this disturbance may represent an excursion from the normal operation, the system still operates within the state of control because all elements are within the predefined criteria. The quality of the output material is not affected, as confirmed by the control strategy and supported by pharmaceutical development data indicating the origin of these disturbances and their potential impact. Therefore, this type of disturbance is not uncommon, no material needs to be diverted, and collection of the output material continues.

3.2. Case #2

Type 2: A transient disturbance which sustains beyond the predefined ranges of duration and/or amplitude (Figure 6) and may impact the quality of the output material.

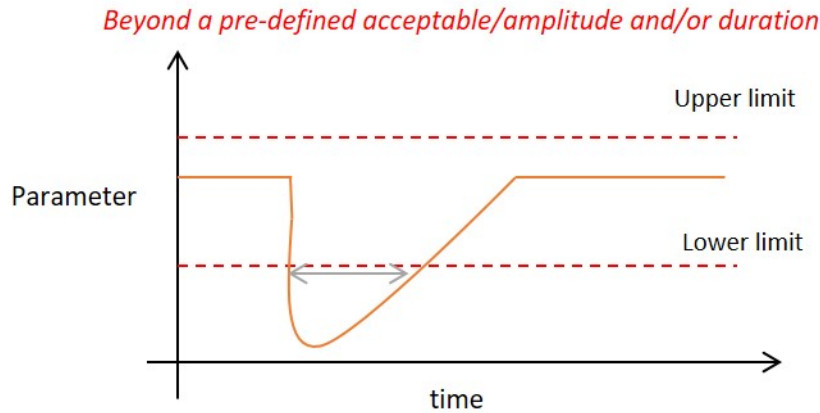


Figure 6: A schematic representation of a Type 2 disturbance

Example: A LIW feeder disturbance with amplitude and/or duration values that require material diversion.

Possible Scenario: One of the feeders encounters flow disruption that causes a reduction in the drug substance concentration (100 to 85% LC for 80 seconds) from the feeder to the blender before returning to 100%. (See Example 2 in Figure 6). All other elements are within the pre-defined criteria.

Impact on Quality: The process model predicts the drug substance concentration within the tablet press feed frame will exceed the predefined acceptance limits of $100 \pm 7\%$. The NIR measurements confirm the model prediction. Based on product and process understanding, the overall magnitude of this disturbance is significant, such that the capacity of the system (mainly blender) is unable to provide an adequate dampening effect to smooth out this sudden transient decrease in the drug substance concentration. It should be noted that the process model was part of the control strategy. If there is a disagreement between the model prediction and the NIR measurement, this may trigger an investigation of decision making and then documentation would follow.

Response: The non-conforming material is diverted based on the preestablished procedure and criteria in the automation system. For example, the time to start and end diversion is controlled by the automation system and considers system dynamics relevant to the NIR location and the point of material diversion. The system returns to normal product collection mode once the non-conforming material is completely diverted. A contemporaneous investigation may be initiated to determine root cause, and the results of this investigation may help inform the next step.

3.3. Case #3

Type 3: Multiple Type 1 disturbances which occur periodically (Figure 7) and may or may not impact the quality of the output material, depending the frequency of occurrence.

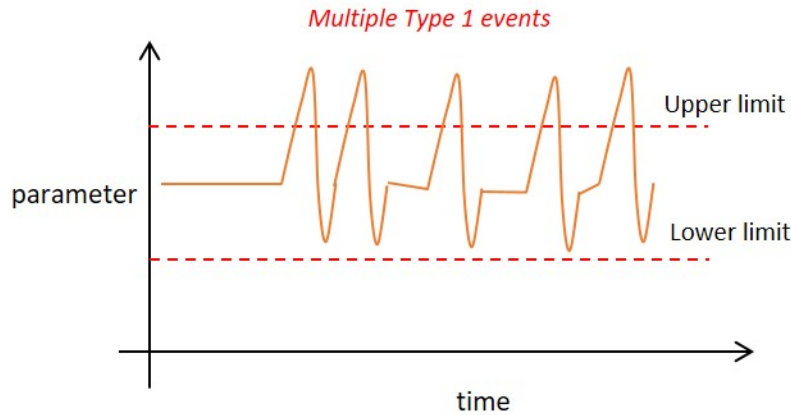


Figure 7: A schematic representation of a Type 3 disturbance

Possible Scenarios: Events such as power surge and mechanical failure can cause multiple Type I disturbances (e.g., feeder disturbance of a few seconds coupled with disturbance in tablet weight and other IPCs). All elements are within the pre-defined criteria.

Impact on Quality: All elements of the control strategy were maintained within the pre-defined criteria. However, if the frequency of Type 1 disturbances is high and these disturbances occur closely over a period of time, the system may not be able to provide a sufficient damping effect on these small but periodic changes, resulting in potentially non-conforming materials. The impact of this type of disturbances needs to be monitored closely by NIR and other elements of the control strategy.

Response: Process and product collection for this case may continue to proceed as normal as long as all control strategy elements meet the predefined criteria. If one or more elements fail to meet the predefined criteria, the affected materials need to be diverted on the preestablished procedure and criteria in the automation system, and the process should be paused or shut down immediately to determine the root cause. Regardless of whether the control elements meet the predefined criteria or not, there is a need to understand if the disturbances are part of normal variation or special cause. This may warrant the assessment of process capability or other evaluations. If this type of disturbance is not due to a special cause, additional efforts to understand and reduce or eliminate these disturbances will be warranted. In such an instance, the process should be stopped until the investigation is completed.